

SECTION 4

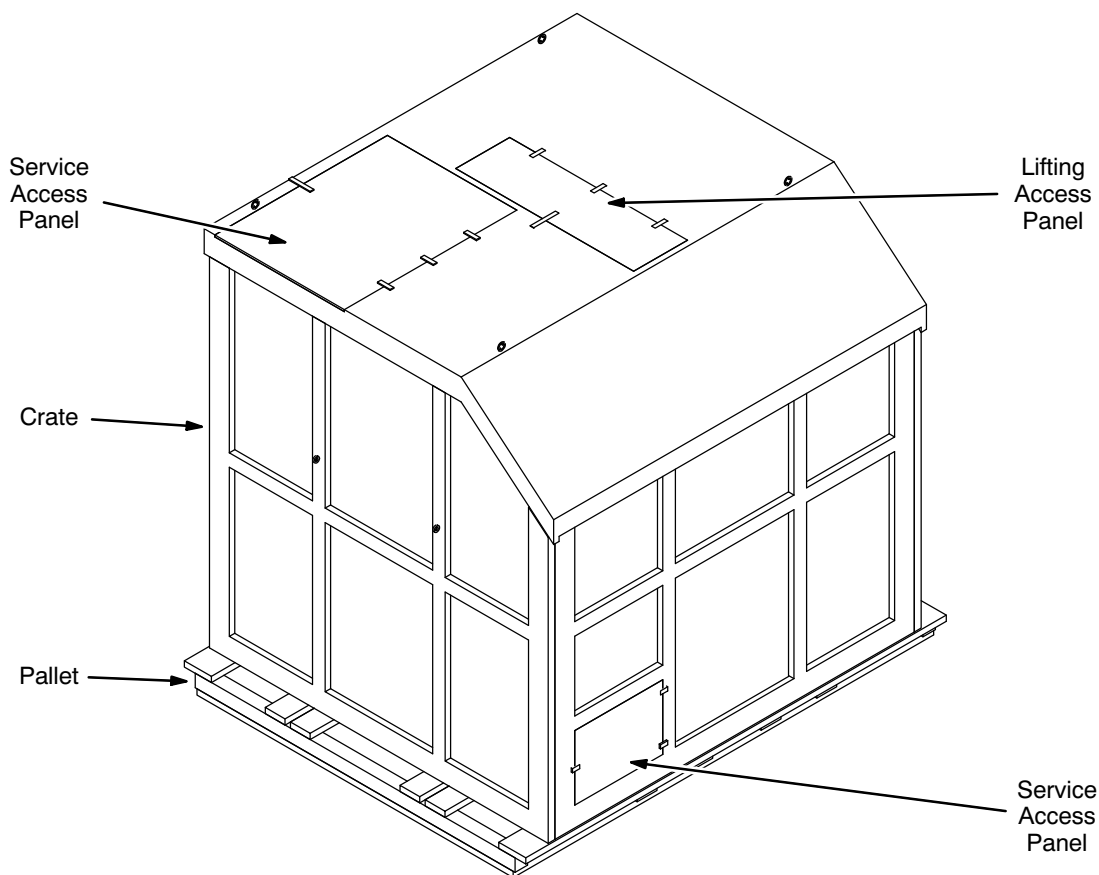
INTERNATIONAL (AIR / SEA) MAGNET DELIVERY

WARNING!

LOSS OF MAGNET VACUUM WILL RESULT IN THE VIOLENT EXPULSION OF HELIUM GAS, WHICH CAN CAUSE ASPHYXIATION IN ENCLOSED AREAS. USE EXTREME CAUTION TO NOT CONTACT / DAMAGE THE VACUUM VESSEL DURING MAGNET TRANSIT OR SITING.

Note

The international shipping crate is shown in Illustration 4-1. Provisions to allow filling and electrical / temperature checks while magnet is still in shipping crate are indentified. These enable required service based upon magnet shipping date to be performed in transit.



CRATE / PALLET
ILLUSTRATION 4-1

4-1 SHIPPING AND HANDLING GUIDELINES

CAUTION

Review guidelines with Shipper / Riggers prior to transporting magnet.

Shipping and handling guidelines are provided in Tables 4-1, 4-2 and 4-3.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

TABLE 4-1
LIFTING / HANDLING INFORMATION

Magnet	Maximum Weight*	Maximum Tilt**	Forklift Capability
GE 0.7T	24,000 lbs. (10,890 kg.)	30 deg.	Only with pallet attached
* Weight includes magnet, poleface, helium, motorshields, gradient coils with cables, hoses, etc. and international shipping crate. ** Maximum tilt angle off vertical axis is allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).			

CAUTION

Make sure conditions specified in Tables 4-2 and 4-3 are met to prevent internal damage to magnet. The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 4-2
SHIPPING INFORMATION

Magnet	Allowable Shipping Modes	Shipping Capability	Maximum Transit Time
GE 0.7T	Airplane, Air ride only trailer, Boat or ocean going ship	Warm or Cold	Cryogen refill for cold shipment ≈ 30 days (to eliminate magnet precool requirement)

TABLE 4-3
MAXIMUM LOAD INFORMATION

Magnet	Maximum Longitudinal Force	Maximum Lateral Force	Maximum Vertical Force
GE 0.7T	1.2 G	1.5 G	1.5 G

4-2 EQUIPMENT / TOOLS**4-2-1 Forklift / Crane**

TABLE 4-4
FORKLIFT / CRANE REQUIREMENTS

Item	Qty.	Capacity / Range	Furnished By:	Intended Use:
Forklift	1	24,000 lbs. (10,890 Kg) at 50 inches (1,270 mm) to load center	Rigger	Unloading or moving magnet with pallet and crate installed
		Minimum fork length: 90 in. (2,286 mm)		
		Minimum distance between forks: 40 in. (1,016 mm)		
Crane	1	24,000 lbs. (10,890 Kg)	Rigger	Unloading or moving magnet
Sling (Strap or Cable)	2	2,000 lbs. (910 Kg) work load limit, 6 ft. (1830 mm) min. length	Rigger	Crate removal
Shackle or hook	2	2,000 lbs. (910 Kg) work load limit	Rigger	Crate removal
Wrench	1	Size: 9/16 in. and 3/8 in.	Rigger	Crate removal

Note

For international shipments lifting straps are already attached to the magnet and are accessible from the top access panel. These straps are to remain attached to the magnet throughout magnet transit.

4-2-2 Miscellaneous Equipment / Tools

TABLE 4-5
MISCELLANEOUS EQUIPMENT / TOOLS FOR MOVING AND CHECKING MAGNET

Item	Qty.	Equipment / Tool Requirement	Responsible	Function
Magnet System Components Checks	1	Direction 2243897, GE 0.7T OpenSpeed Magnet Delivery and Installation	GE / SYS	Confirm electrical check values
Hydraulic or Toe Jack	4	Must support one end of magnet on 2 jacks or both ends of magnet on 4 jacks	Rigger	Raise up magnet for roller dollies or leveling plates
		Carrying capacity: 12,000 lbs. (5,450 Kg) per jack		
Roller Dollies	2	Carrying capacity: 6,000 lbs. (2,722 Kg) per dolly	Rigger	Moving magnet to Magnet Room
Come-Along	1	Pulling capacity: 12,000 lbs. (5,443 Kg)	Rigger	Pull magnet into position
Level	1	24 – 36 in. (600 – 1000 mm) long	Rigger	Check magnet level
Torque Wrench and Socket	1	Torque range = 58 to 258 ft.-lbs.; Deep socket, size dependant on anchor selection	Rigger	Bolt down magnet
Socket Wrench and Sockets	1	3/4 in. socket size (add sockets for crate removal)	Rigger	Remove 1/2 in. crate lag screws
Portable Temperature Meter (2125073)	1	N / A	GE / FE	2125073 Check internal temperatures
Digital Volt / Ohm Meter	1	Fluke model 87 or equivalent	GE / FE	Perform magnet electrical checks
Wood Timber	2	4 in. x 6 in. x 8 ft. (100 x 200 x 2000 mm)	Rigger	Moving / positioning magnet to Magnet Room
Steel or Aluminum Plate	6	.25 in. x 18 in. x 6 ft. (7 x 500 x 1800 mm)	Rigger	Moving magnet to Magnet Room
Pry-Bar	2	Length as required to move magnet	Rigger	Moving / positioning magnet in Magnet Room
RuO Temperature Meter (2171219)	1	N / A	GE	Check second stage & Recondenser temperatures
Adapter Cable for Diode Temperature Meter (46-301620P1)	1	N / A	GE	Check internal temperatures
Circular or Chain Saw	AR	N / A	Rigger	For cutting wood timbers
Heavy Duty Drill & Drill Bits	AR	N / A	Rigger	To add holes to wood timbers
Min. .50 Inch Dia. Mounting Hardware	AR	N / A	Rigger	For attaching wood timbers to rails

4-3 IN-TRANSIT HANDLING AND SERVICE

4-3-1 In-Transit Helium Fill

Cryogen depletion occurs in approximately 3 days after coldhead power is removed from a full magnet. The magnet warms up to liquid nitrogen temperature (77 K) in approximately 5.5 days. If the coldheads are off for an extended period of time (> 5 days) due to transit, storage or power outage, refilling the magnet to keep it cold should be considered to eliminate the need to precool it before filling later. Perform helium fill in conformance with magnet service manual Direction 2231934, SET-UP AND CALIBRATION, Section 3-4, Liquid Helium Fill.

Note

The service access panel on top of the shipping crate must be opened to refill magnet with liquid helium. See Illustration 4-1.

WARNING!

- **THIS PROCEDURE MUST BE PERFORMED BY QUALIFIED PERSONNEL ONLY. MAKE SURE THE SAFETY PRECAUTIONS LISTED BELOW ARE COMPLETELY FOLLOWED WHEN PERFORMING ANY SERVICE THAT WILL RESULT IN HELIUM EXHAUST FROM THE MAGNET.**
- **LARGE QUANTITIES OF HELIUM GAS WILL DISPLACE OXYGEN IN A CLOSED ROOM. GOOD VENTILATION IS REQUIRED FOR YOUR PERSONAL SAFETY. MAKE SURE THE MAGNET STORAGE AREA IS ADEQUATELY VENTILATED BY OPENING DOORS AND WINDOWS. THE ADDITION OF PORTABLE FANS MAY BE NECESSARY IN CASES OF A SMALL-SIZE ROOM WITH POOR VENTILATION.**
- **SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING, GLOVES (NONABSORBENT MATERIAL) AND GOGGLES WHEN TRANSFERRING CRYOGENS. MAKE SURE A SECOND PERSON IS PRESENT WHILE SERVICING THE MAGNET IN CASE OF ANY EMERGENCY.**
- **SMOKING IS PROHIBITED IN THE MAGNET AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN THUS PRODUCING A HIGHLY ENRICHED OXYGEN LIQUID, WHICH IS HIGHLY COMBUSTABLE.**

■ 4-3-2 In-Transit Handling with Crane

WARNING!

DO NOT STAND OR PUT FULL BODY WEIGHT ON TOP OF CRATE TO PREVENT FALLING OFF CRATE OR THROUGH CRATE ROOF.

- 1. Position hook / spreader beam of crane centrally over crate. Refer to Table 4-4 for crane capacity specification.
- 2. Open hatch labelled “LIFTING ACCESS” on top of crate, revealing two lifting straps attached to lifting shackles on the magnet. See Illustration 4-2.
3. Attach the lifting straps to the crane hook / spreader beam in conformance with Illustration 3-3.

CAUTION

Carefully control magnet / crate / pallet package while lifting, moving or lowering. Some rotation / tilting will occur. Lower smoothly. Avoid impacts, jolts or excessive tilting (as specified in Table 4-1) as magnet damage may result.

- 4. Carefully lift the entire magnet / crate / pallet package, controlling the tilt and rotation that will occur. Avoid impacts, jolts or excessive tilting (as specified in Table 4-1). When lowering, ease entire package to rest on a flat, smooth surface. Do not rest magnet / crate / pallet any place that is not flat nor strong enough to support 24,000 pounds (10,087 kg).

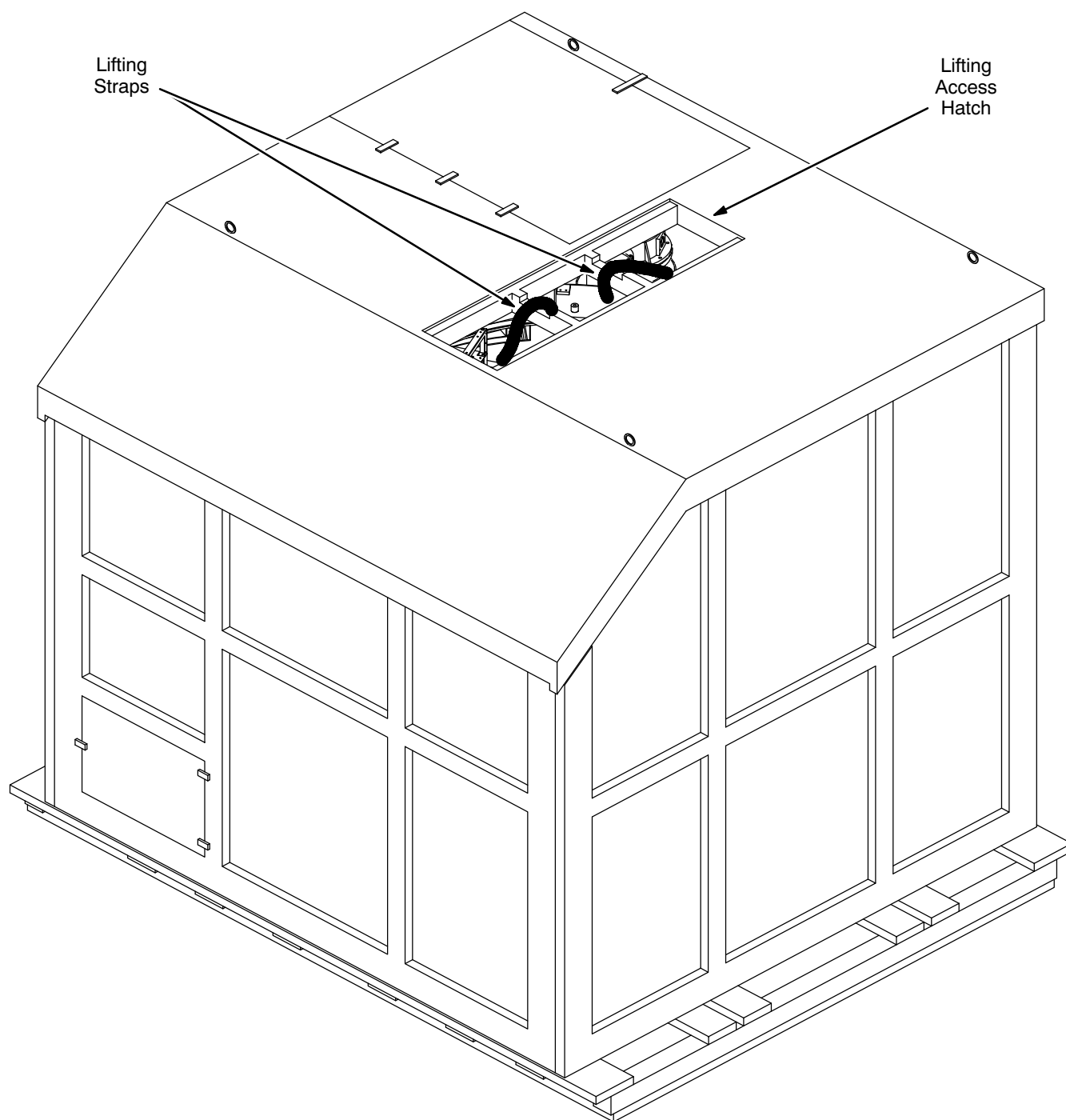
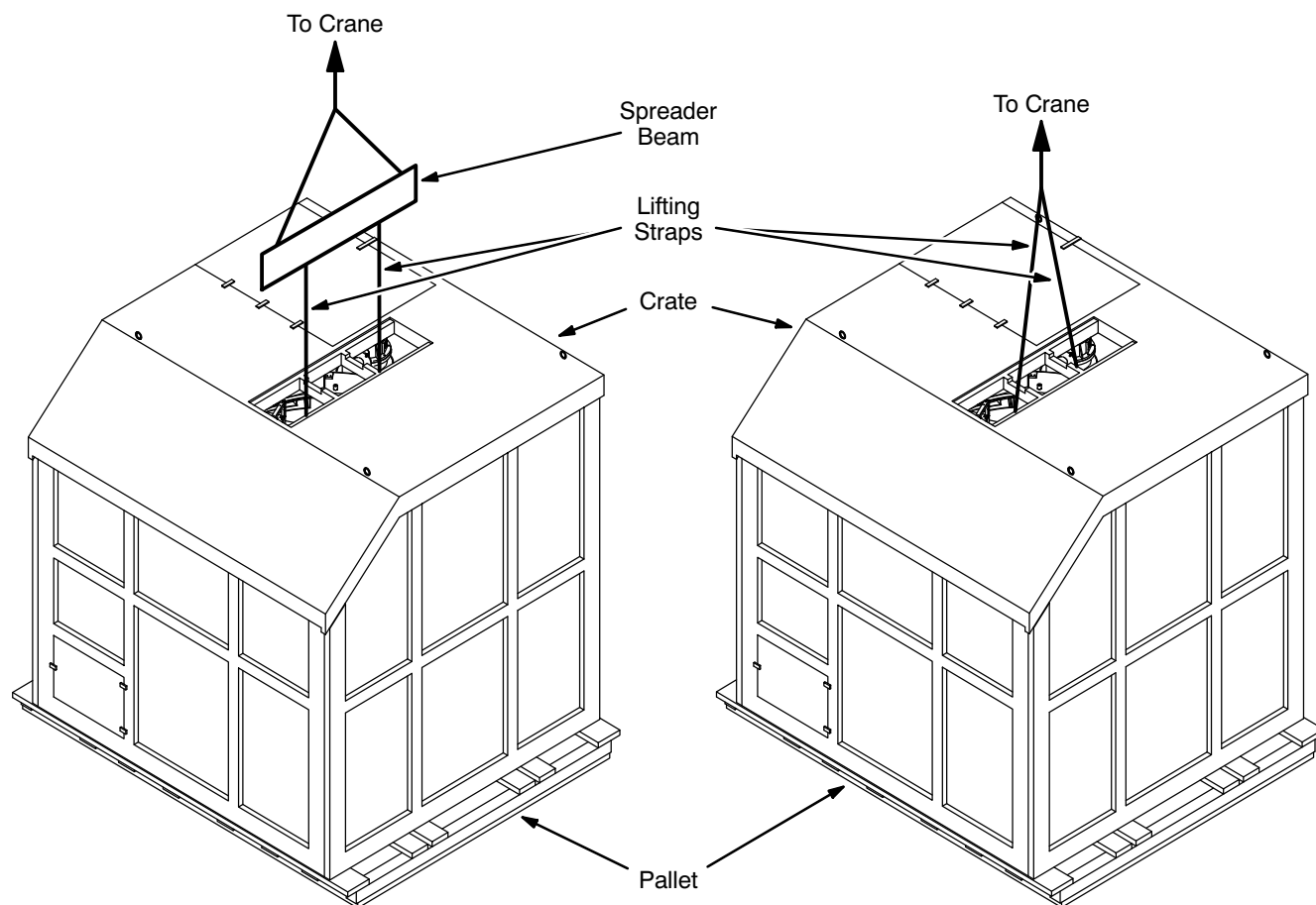
4-3-2 In-Transit Handling with Crane (continued)**LIFTING ACCESS HATCH AND LIFTING STRAPS**

ILLUSTRATION 4-2

4-3-2 In-Transit Handling with Crane (continued)

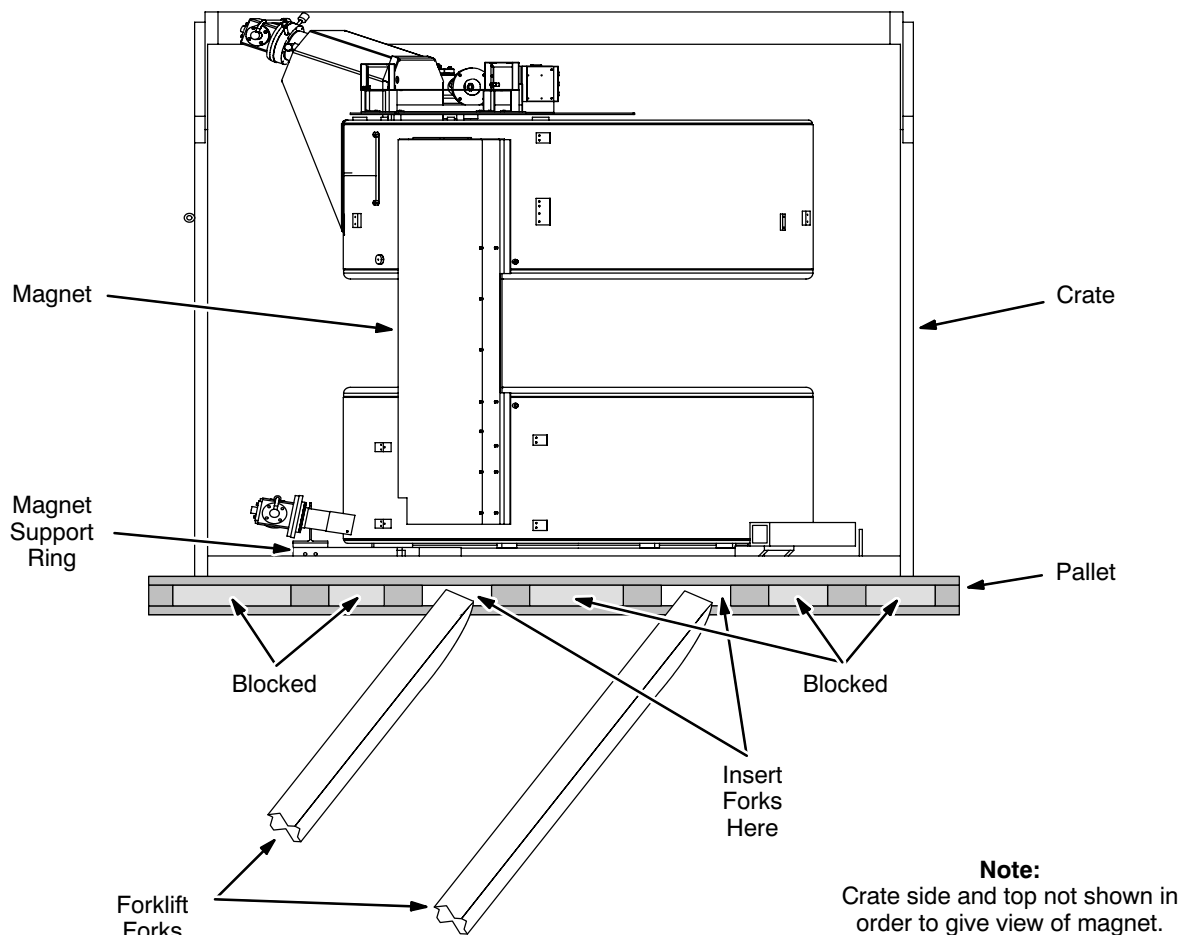
CRANE LIFTING OF CRATE IN-TRANSIT
ILLUSTRATION 4-3

4-3-3 In-Transit Handling with Forklift

CAUTION

It is important to use the proper size forklift and to lift the entire magnet, crate and pallet smoothly to avoid impacts, jolts or excessive tilting (as specified in Table 4-1) which may cause damage to the magnet. Extreme care must be exercised during forklift operations. The magnet pallet must be picked up from the sides only. The forks **MUST** be fully inserted into correct openings under shipping pallet. Make sure the forklift capacity and fork dimensions meet the specifications stated in Table 4-4.

1. Locate the forklift openings on the side of the shipping pallet. See Illustration 4-4.
2. Extreme care must be exercised during forklift operations. The magnet / crate / pallet package must be picked up from the sides only. Position the forks at the forklift openings on side of pallet and insert carefully. See Illustration 4-4.



LOCATION OF FORKLIFT OPENINGS IN SHIPPING PALLET

ILLUSTRATION 4-4

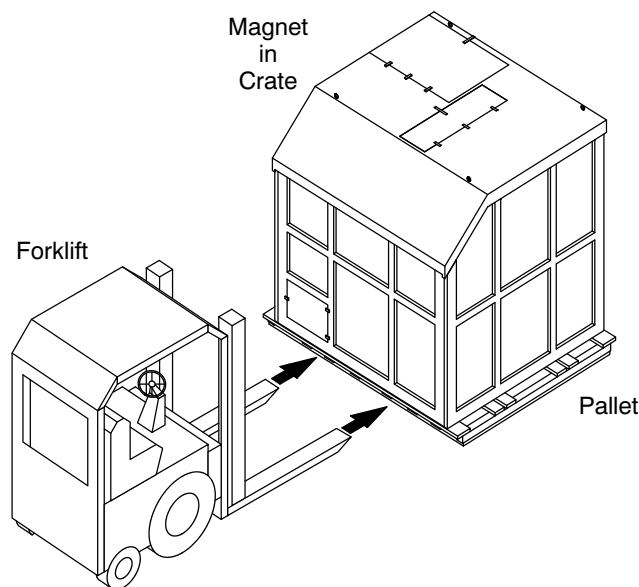
4-3-3 In-Transit Handling with Forklift (continued)

3. Carefully drive the forklift until forks are completely inserted into the pallet openings.

CAUTION

Carefully lift, move and lower magnet / crate / pallet package to avoid impacts, jolts or excessive tilting (as specified in Table 4-1) as magnet damage may result.

4. Carefully lift the magnet / crate / pallet package and move to desired location. Avoid impacts, jars or excessive tilting (as specified in Table 4-1) to crate / pallet as magnet damage may result. When lowering, ease entire package to rest on a flat, smooth surface strong enough to support 24,000 pounds (10,087 kg).



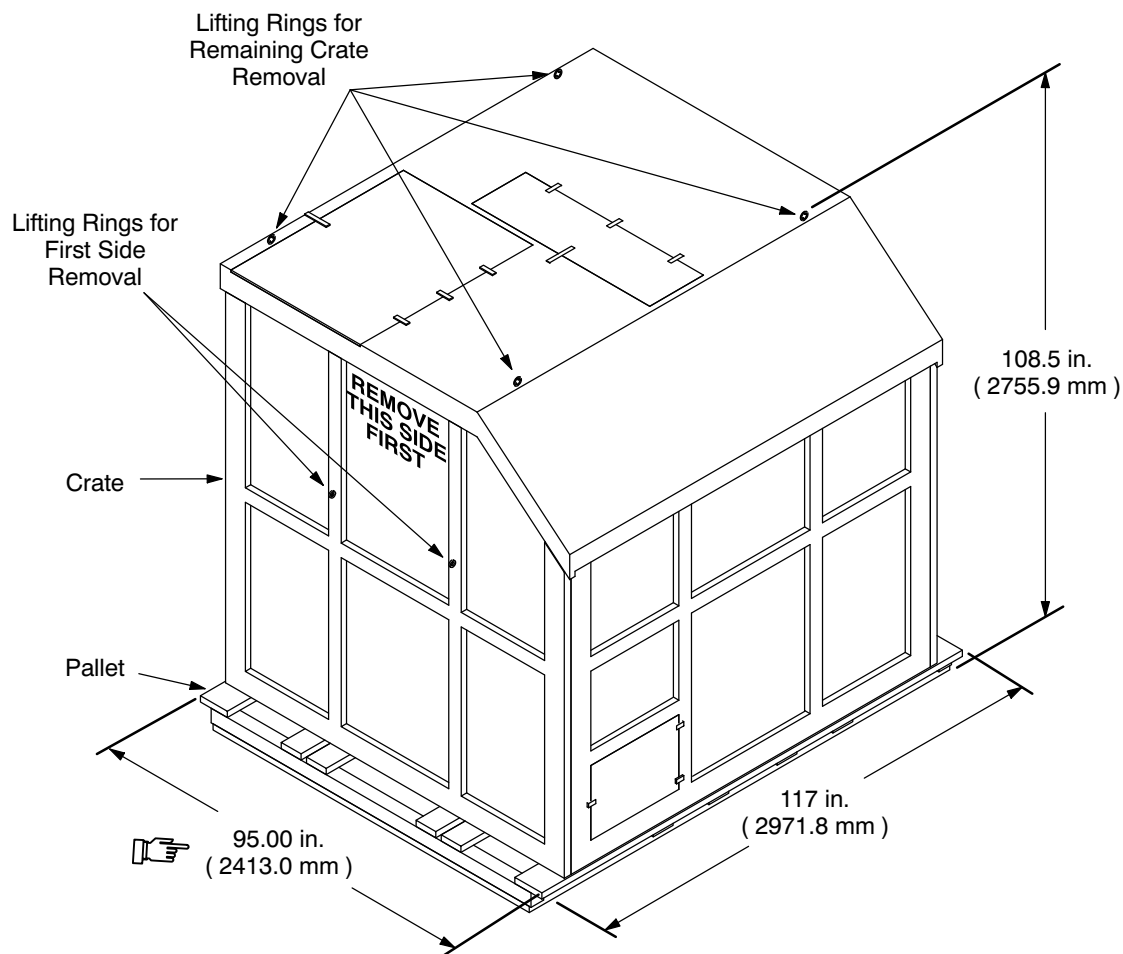
LIFTING CONFIGURATION FOR FORKLIFT
ILLUSTRATION 4-5

4-4 CRATE REMOVAL

IMPORTANT !!!

During this procedure carefully inspect all packaging and their contents for damage that may have occurred during shipping. If any damage is found, follow the procedure found at the front of this manual under DAMAGE IN TRANSPORTATION.

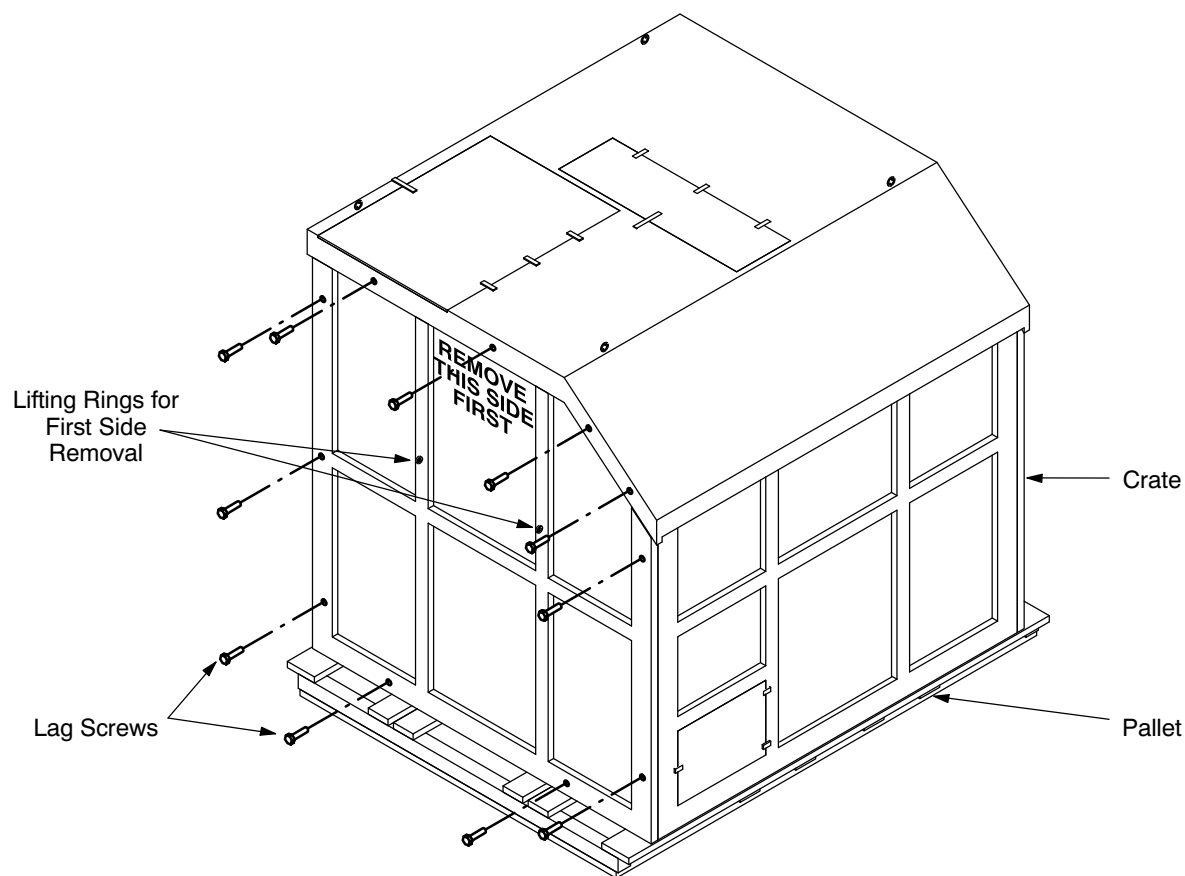
1. Remove all subsystem crates, except the magnet crate, from trailer / transport using a crane or forklift. Inspect all crates for visible damage. Report any damage found according to the "Damage in Transportation" procedure at the front of this manual.
2. Move subsystem crates to a receiving location protected from the weather, preferably close to and at the same level as the examination room.
3. Inspect the crate containing the magnet and identify the crate side marked "REMOVE THIS SIDE FIRST". This designated side is to be removed first. See Illustration 4-6.



INTERNATIONAL SHIPMENT CRATE
ILLUSTRATION 4-6

4-4 CRATE REMOVAL (continued)

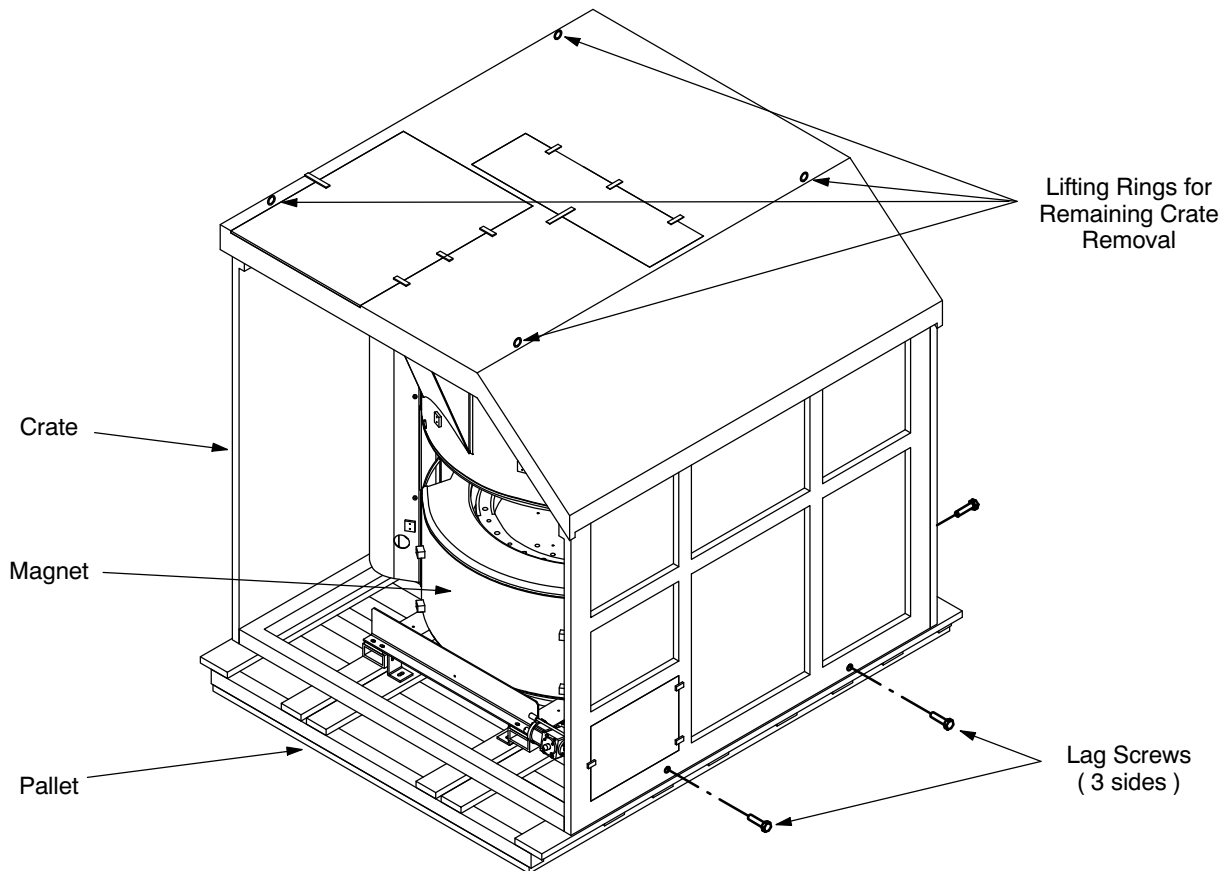
4. Position crane centrally over the side of the crate.
5. Connect the crane to the two lifting rings located on the “REMOVE THIS SIDE FIRST” side of the crate. Use straps / cables and shackles supplied by riggers. Tighten the straps / cable snug.
6. Remove lag screws securing the face of the crate marked “REMOVE THIS SIDE FIRST” See Illustration 4-7.
7. Lift the “REMOVE THIS SIDE FIRST” side of crate with crane and clear from the area.



LAG SCREW REMOVAL, FIRST SIDE
ILLUSTRATION 4-7

4-4 CRATE REMOVAL (continued)

8. Remove the lag screws securing the bottoms of the three remaining sides of crate to pallet. See Illustration 4-8.

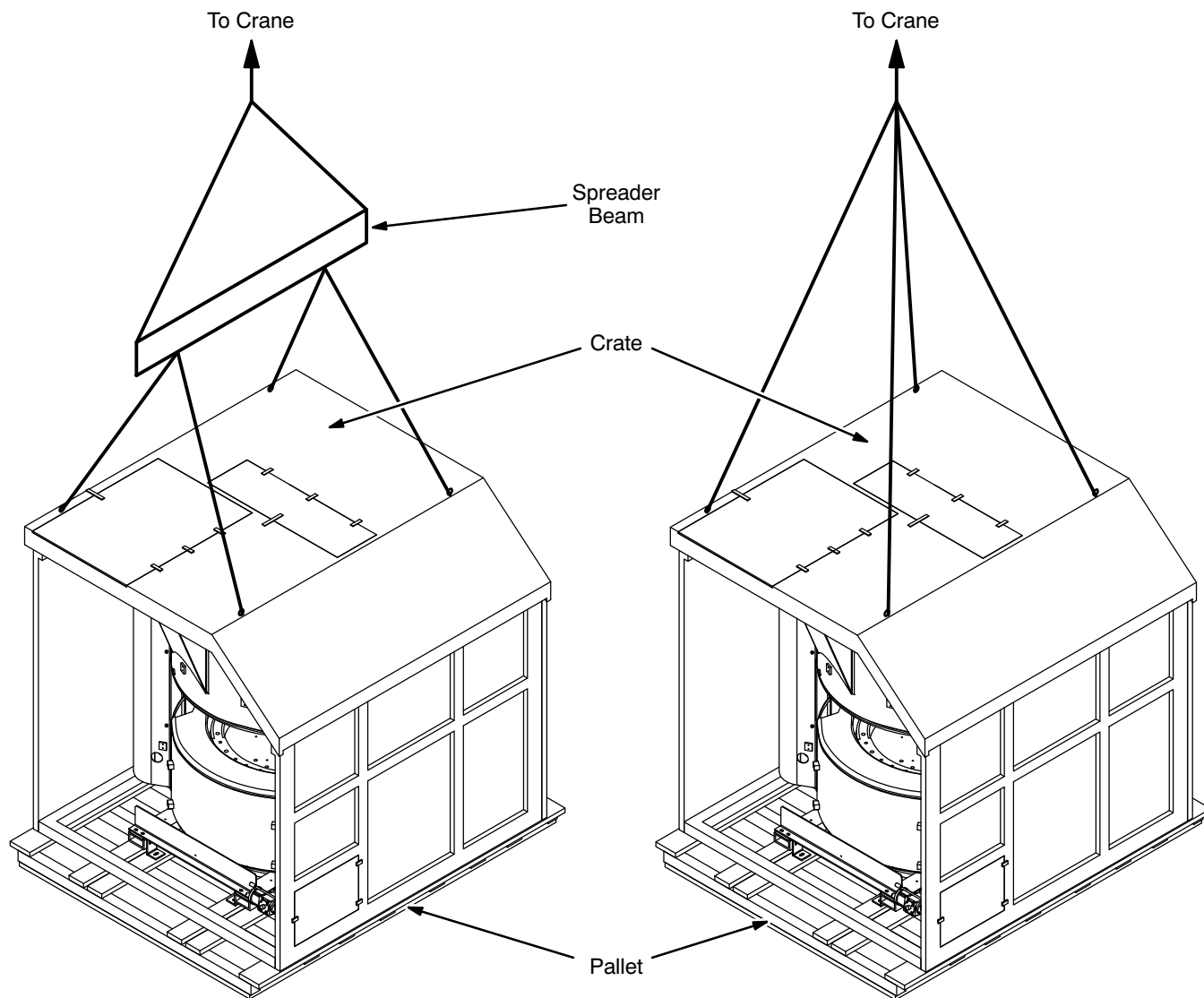


LAG SCREW REMOVAL, REMAINING SIDES
ILLUSTRATION 4-8

9. Position crane's hook / spreader beam centrally over crate. Refer to Table 4-4 for minimum crane specification.
10. Secure four straps / cables of one-ton (907 Kg) working load capacity each and a minimum six feet (1829 mm) long to the four lifting rings on the crate's top side. Use a one-ton (907 Kg) anchor shackle on each sling.
11. Attach the straps / cables to crane's hook / spreader beam. See Illustration 4-9.
12. Carefully lift crate sides and top straight up and clear of the magnet using crane. Do not let crate scrape or hit magnet. Swing away to remove.

Note

Do not remove the protective plastic bag / wrap surrounding magnet until required for installation purposes inside the Magnet Room.

4-4 CRATE REMOVAL (continued)

CRAN LIFTING CRATE TO REMOVE CRATE TOP
ILLUSTRATION 4-9

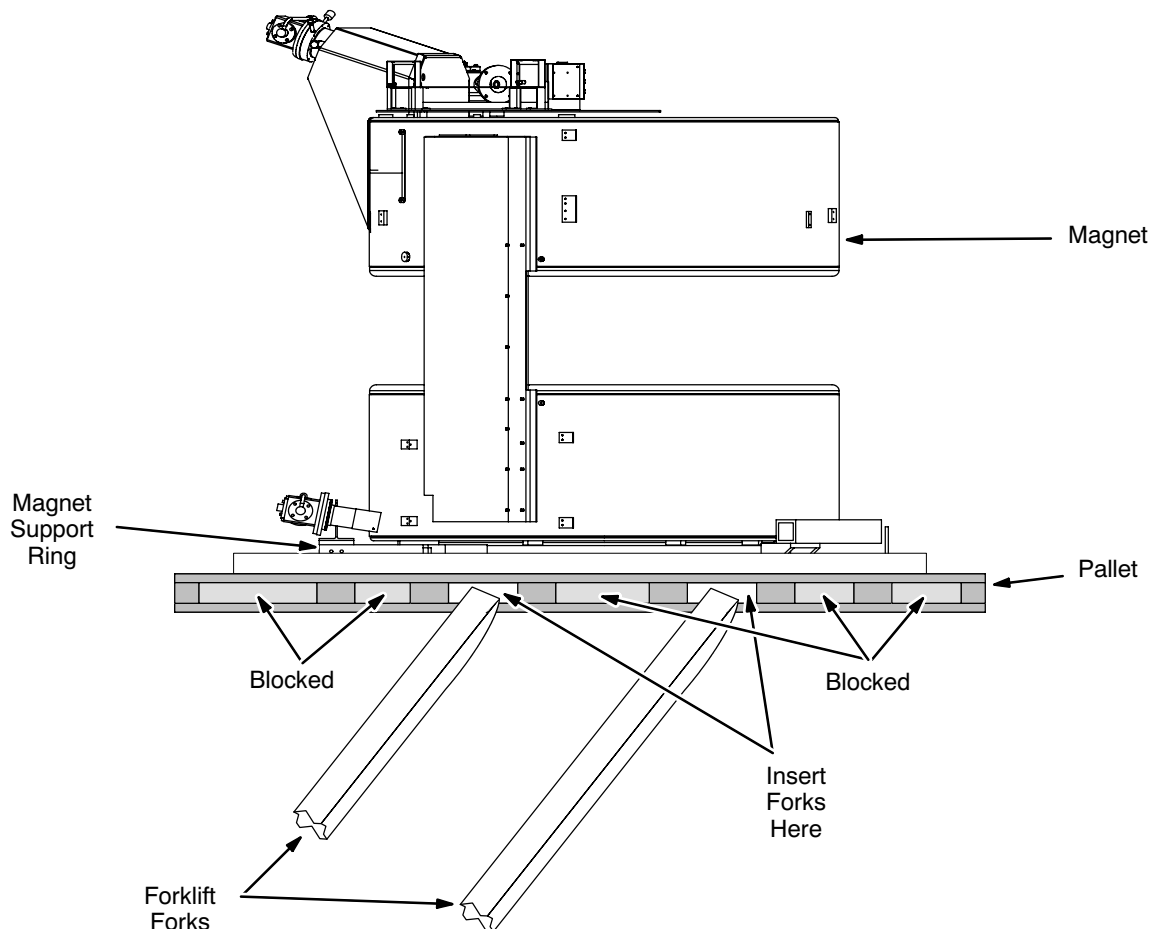
4-5 UNLOADING MAGNET AND PALLET

4-5-1 Forklift Method of Unloading

CAUTION

It is important to use the proper size forklift and to lift the magnet / pallet smoothly to avoid impacts, jolts or excessive tilting (as specified in Table 4-1) which may cause damage to the magnet. Extreme care must be exercised during forklift operations. The magnet pallet must be picked up from the sides only. The forks **MUST** be fully inserted into correct openings under shipping pallet. Make sure the forklift capacity and fork dimensions meet the specifications stated in Table 4-4.

1. Locate the forklift openings on the side of the shipping pallet. See Illustration 4-10.



LOCATION OF FORKLIFT OPENINGS IN SHIPPING PALLET

ILLUSTRATION 4-10

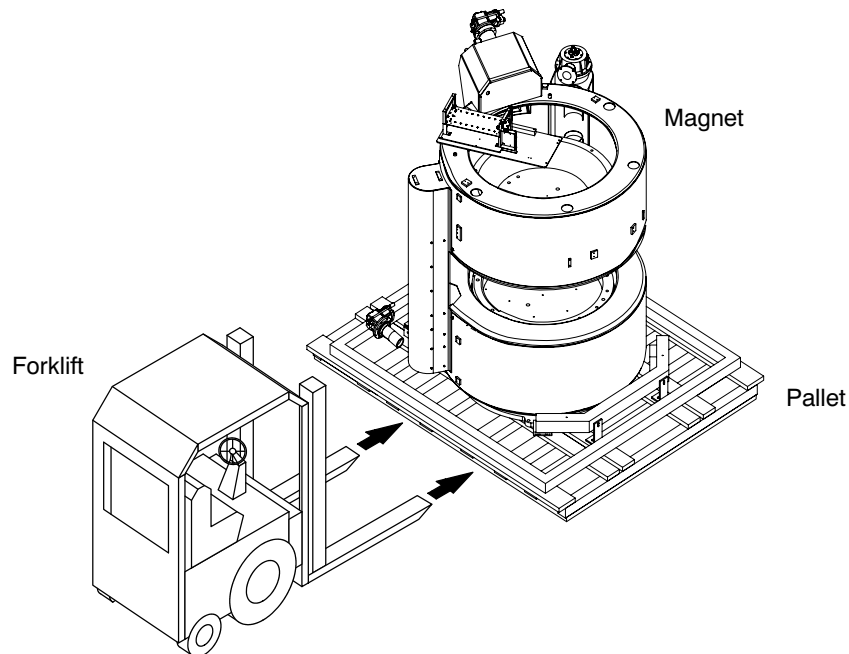
4-5-1 Forklift Method of Unloading (continued)

2. Extreme care must be exercised during forklift operations. The magnet / pallet must be picked up from the sides only. The forks **MUST** be placed directly under the four (4) brackets attaching magnet to shipping pallet. Position the forks at the forklift openings on side of pallet. See Illustration 4-10.
3. Carefully drive the forklift until forks are completely inserted into the pallet openings, as in Illustration 4-11.



Carefully lift, move and lower magnet / crate to avoid impacts, jolts or excessive tilting (as specified in Table 4-1) as magnet damage may result.

4. Lift magnet / pallet with forklift and move smoothly to desired location. Avoid impact or jars to magnet. When lowering, ease magnet and pallet to rest on a flat, smooth surface.
5. After unloading magnet and pallet, removal of pallet will require a crane.



LIFTING CONFIGURATION, FORKLIFT TO MAGNET ON PALLET
ILLUSTRATION 4-11

■ 4-5-2 Crane Method of Unloading

CAUTION

Pallet MUST be removed from magnet prior to any magnet lift operations, to allow for magnet transport / moving using dollies / jacks.

1. Unbolt and remove the four 1 inch (25.4 mm) nuts which secure the pallet to the magnet. See Illustration 4-12. For rigging / lifting details, refer to Section 3, Domestic Magnet Delivery.

Note

Do not remove the protective plastic bag / wrap surrounding magnet until required for installation purposes inside the Magnet Room.

2. Position crane hook / spreader beam centrally over magnet for a vertical lift. See Illustration 4-13.
3. Attach straps from lifting shackles on top of magnet to the crane hook / spreader beam in conformance with Illustrations 4-13.

CAUTION

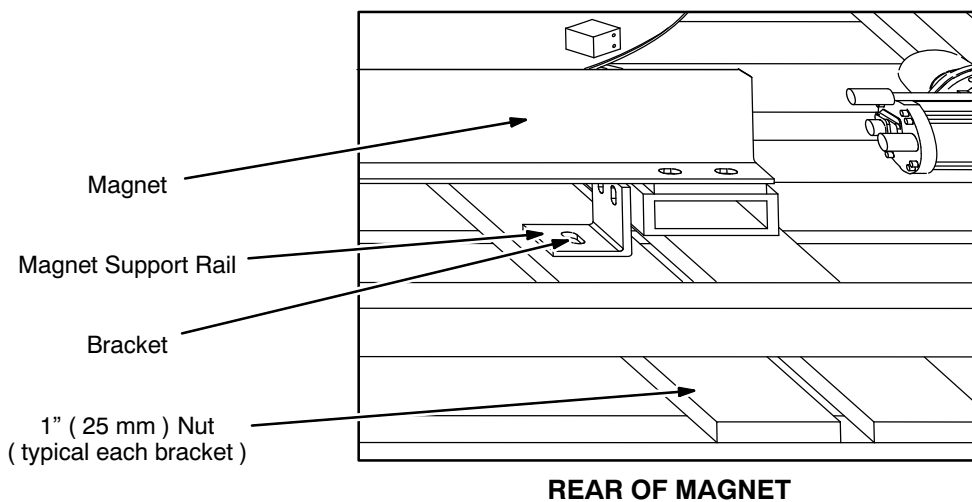
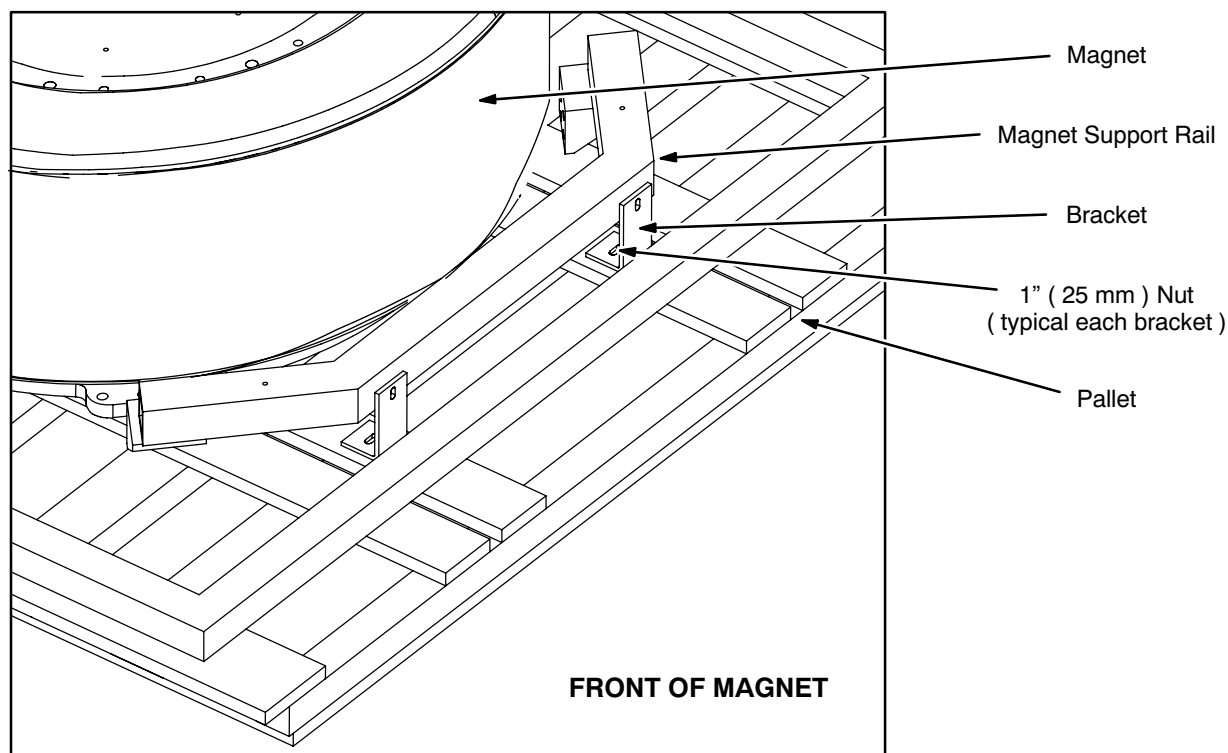
Carefully control magnet while lifting, moving or lowering. Some rotation / tilting will occur. Lower smoothly onto a flat surface. Avoid impacts, jolts or excessive tilting (as specified in Table 4-1) as magnet damage may result.

Note

Magnet must be lifted in conformance with requirements shown in Illustrations 4-13 to avoid damage to external components.

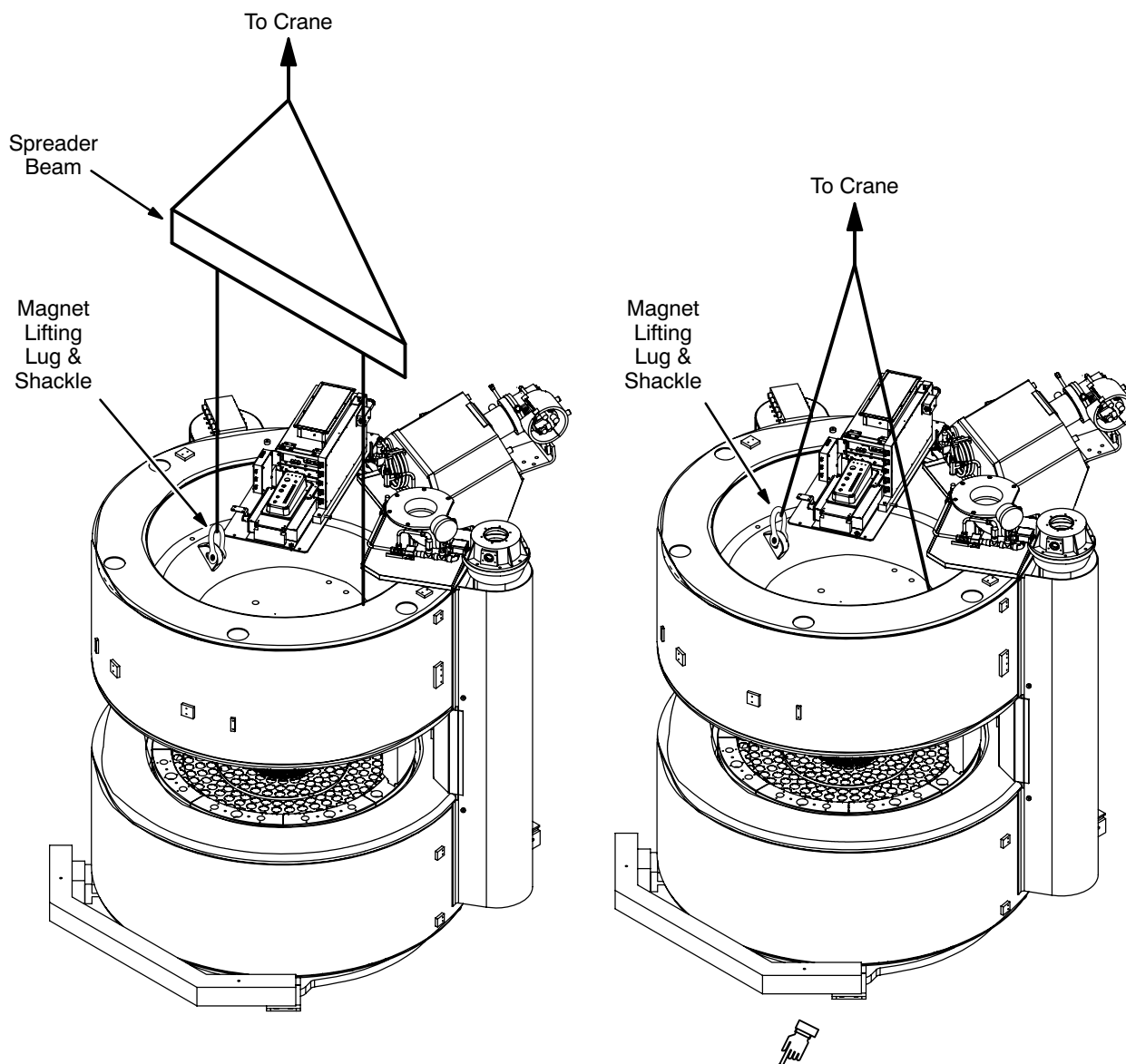
4. Lift magnet. When lowering, ease magnet to rest on a flat, smooth surface.

4-5-2 Crane Method of Unloading (continued)



MAGNET MOUNTING ON SHIPPING PALLET

ILLUSTRATION 4-12

4-5-2 Crane Method of Unloading (continued)

- Note:** 1. Use Lifting Straps preattached to lifting lugs at 35.50 In. (901.7 mm) apart.
2. DO NOT angle lifting straps / cables outwards from magnet as component damage may result.

CRANE LIFTING MAGNET
ILLUSTRATION 4-13